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Distinct Activation Mechanisms of CXCR4 and ACKR3 Revealed by Single-Molecule Analysis of their Conformational Landscapes

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eLife Assessment

This manuscript describes the characterization of the conformational dynamics of two chemokine receptors at the single-molecule level using FRET. The authors make a **convincing** case for attributing the distinct interaction and pharmacology of the two receptors to differences in their conformational energy landscape. These **important** findings will be of interest to scientists working on activation mechanisms of GPCRs and signal transduction.

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Abstract

The canonical chemokine receptor CXCR4 and atypical receptor ACKR3 both respond to CXCL12 but induce different effector responses to regulate cell migration. While CXCR4 couples to G proteins and directly promotes cell migration, ACKR3 is G protein-independent and scavenges CXCL12 to regulate extracellular chemokine levels and maintain CXCR4 responsiveness, thereby indirectly influencing migration. The receptors also have distinct activation requirements. CXCR4 only responds to wild-type CXCL12 and is sensitive to mutation of the chemokine. By contrast, ACKR3 recruits GPCR kinases (GRKs) and β-arrestins and promiscuously responds to CXCL12, CXCL12 variants, other peptides and proteins, and is relatively insensitive to mutation. To investigate the role of conformational dynamics in the distinct pharmacological behaviors of CXCR4 and ACKR3, we employed single-molecule FRET to track discrete conformational states of the receptors in real-time. The data revealed that apo-CXCR4 preferentially populates a high-FRET inactive state, while apo-ACKR3 shows little conformational preference and high transition probabilities among multiple inactive, intermediate and active conformations, consistent with its propensity for activation. Multiple active-like ACKR3 conformations are populated in response to agonists, compared to the single CXCR4 active-state. This and the markedly different conformational landscapes of the receptors suggest that activation of ACKR3 may be achieved by a broader distribution of conformational states than CXCR4. Much of the conformational heterogeneity of ACKR3 is

linked to a single residue that differs between ACKR3 and CXCR4. The dynamic properties of ACKR3 may underly its inability to form productive interactions with G proteins that would drive canonical GPCR signaling.

Introduction

CXC chemokine receptor 4 (CXCR4) is one of the most intensively studied chemokine receptors due to its central role in driving cell migration during development and immune responses, and in cancer where it promotes tumor growth and metastasis (1–4). As a class A G protein-coupled receptor (GPCR), CXCR4 activates inhibitory Gαi protein signaling pathways to directly control cell movement in response to the chemokine CXCL12 (5). The activated receptor is also phosphorylated by GPCR kinases (GRKs), which promotes arrestin recruitment and cessation of the G protein signal. CXCR4 often works together with atypical chemokine receptor 3 (ACKR3, formerly CXCR7) which indirectly influences migration by scavenging CXCL12 to regulate available extracellular levels of the agonist and in turn the responsiveness of CXCR4 (6–8). In the absence of ACKR3 scavenging, excessive CXCL12 stimulation of CXCR4 leads to downregulation, resulting in profound effects on neuronal cell migration and development (9–11).

In contrast to CXCR4 and with some exceptions noted (12, 13), ACKR3 lacks G protein activity and instead is considered to be arrestin-biased (14). However, we and others (9) have shown that arrestins are dispensable for chemokine scavenging while GRK phosphorylation is critical for this function (15), suggesting that ACKR3 might be better described as GRK-biased. The molecular basis for its inability to couple to G proteins in most cell types remains an unanswered question. Our recently determined structures of ACKR3-ligand complexes showed the expected hallmarks of GPCR activation including “microswitch residues” in active state configurations, displacement of transmembrane helix 6 (TM6) away from the helical bundle and an open intracellular pocket, consistent with a receptor that should be able to activate G proteins (16). Accordingly, we replaced the intracellular loops (ICLs) of ACKR3 with those of CXCR2, a canonical G protein-coupled chemokine receptor; however, these changes did not lead to G protein activation (16), suggesting that the lack of coupling is not due to the absence of specific residue interactions. CXCL12 also adopts a distinct pose when bound to ACKR3 compared to CXCR4 and all other chemokines in chemokine-receptor complexes, but since small molecules induce similar biased effector responses, the chemokine pose cannot explain the effector-coupling bias (16). Having excluded other mechanisms we therefore surmised that the inability of ACKR3 to activate G proteins may be due to differences in receptor dynamics. Consistent with this hypothesis, the ICLs observed in ACKR3-agonist complexes are disordered, which may preclude productive effector coupling. The dynamic nature of ACKR3 is also suggested by its considerable constitutive activity in recruiting β-arrestins and its high level of constitutive internalization (16–18).

In addition to their distinct effector interactions, ACKR3 and CXCR4 have dramatically different susceptibilities to activation by different ligands. CXCR4 is activated by a single chemokine agonist, CXCL12. Moreover, modifications of the CXCL12 N-terminal signaling domain (e.g. single point mutations as in $\text{CXCL12}_{\text{P2G}}$ or multiple mutations as in $\text{CXCL12}_{\text{LRHQ}}$) transform the chemokine into an antagonist of CXCR4 (19, 20). Mutational analysis, modeling, and new structures of CXCR4 suggest that ligand activation involves a precise network of interacting residues that stabilize the active receptor conformation (21–25). By contrast, ACKR3 is activated by CXCL12 as well as its variants (including $\text{CXCL12}_{\text{P2G}}$ and $\text{CXCL12}_{\text{LRHQ}}$ (19, 20)), other chemokines (CXCL11) (26), other proteins (adrenomedullin, BAM22) (27, 28), and opioid peptides (29). In fact, most ligands for ACKR3 act as agonists, which is best explained by a non-specific “distortion” mechanism of activation whereby any ligand that breaches the binding pocket causes helical movements that are permissive to GRK phosphorylation and arrestin recruitment. A distortion mechanism is consistent with the different binding poses observed for a small molecule agonist compared to the CXCL12 N-terminus in the receptor orthosteric pocket (16), and ligand bulk rather than specific interactions between agonist and ACKR3 being required for activation. We hypothesize that this distortion mechanism would also be facilitated by a receptor that is conformationally dynamic, by enabling nonspecific ligand interactions and more than a single conformation to promote activation.

To investigate the role of conformational dynamics in the distinct pharmacological behaviors of ACKR3 and CXCR4, we developed a single-molecule Förster resonance energy transfer (smFRET) approach. Many ensemble methods such as EPR, NMR and fluorescence-based methods have provided considerable insights into GPCR dynamics, conformational heterogeneity, and exchange between distinct structural states (30–36). However, these methods are limited in their ability to resolve the sequence of state-to-state transitions except in rare cases where transitions can be temporally coordinated with high precision (37). By contrast smFRET enables detection of sparsely populated states, reveals the sequence of state-to-state transitions and provides kinetic information through analysis of state dwell times. For example, smFRET studies of the β_2 adrenergic receptor (β_2AR) revealed a dynamic equilibrium between inactive and active conformations that was responsive to agonist and G protein binding (38). More recent smFRET studies of the glucagon receptor (39) and the A_{2A} receptor (A_{2A}R) (40) have documented the existence of distinct intermediate conformations in addition to inactive and active receptor conformations.

Here we present the first smFRET study of the chemokine receptors CXCR4 and ACKR3. Our experimental system allows real-time observation of the conformational fluctuations of individual receptor molecules in a native-like lipid environment and assessment of the differences in the conformational dynamics of the two receptors in their apo states and in response to ligands. Our results indicate that ACKR3 is more dynamic and conformationally heterogeneous than CXCR4 or other class A GPCRs previously studied, which may explain its activation prone nature and lack of G protein coupling. In contrast, CXCR4 appears less flexible, consistent with a more restricted, structurally-defined activation mechanism. Together these data characterize the molecular differences between CXCR4 and ACKR3 and suggest that enhanced conformational dynamics plays an important role in the atypical function of ACKR3.

Results

Development of smFRET experimental systems for CXCR4 and ACKR3

To visualize conformational fluctuations of CXCR4 and ACKR3 by smFRET, cysteine residues were introduced into CXCR4 at positions $150^{4.40}$ in TM4 and $233^{6.29}$ in TM6, and at $159^{4.40}$ and $245^{6.28}$ in ACKR3 (Figs. S1A & B) (numbers in superscript refer to the Ballesteros-Weinstein numbering scheme for GPCRs) for covalent labeling with FRET donor (D, Alexa Fluor 555, A555) and acceptor (A, Cyanine5, Cy5) fluorophores. Single receptor molecules of labeled CXCR4 or ACKR3 were reconstituted into phospholipid nanodiscs to mimic the native membrane bilayer environment. Nanodiscs have been used in a wide variety of structural and biophysical studies of integral membrane proteins, including ACKR3, and reconstitute protein-lipid interactions lost in detergent systems (41–43). The nanodisc-receptor complexes were then tethered to a quartz slide through biotinylation of the nanodisc membrane scaffolding protein (MSP) (**Fig. 1A**). To promote monomeric receptor incorporation in each nanodisc, we utilized MSP1E3D1, which forms nanodiscs with a diameter of approximately 13 nm (44, 45).

Crystal and cryo-electron microscopy (cryo-EM) structures of homologous class A GPCRs, such as β_2AR, in inactive and active conformations, reveal that TM6 moves outwards from the TM helical bundle during activation, whereas the position of TM4 remains relatively fixed (46). Accordingly, we anticipated that labeling at the positions indicated above would be sensitive to

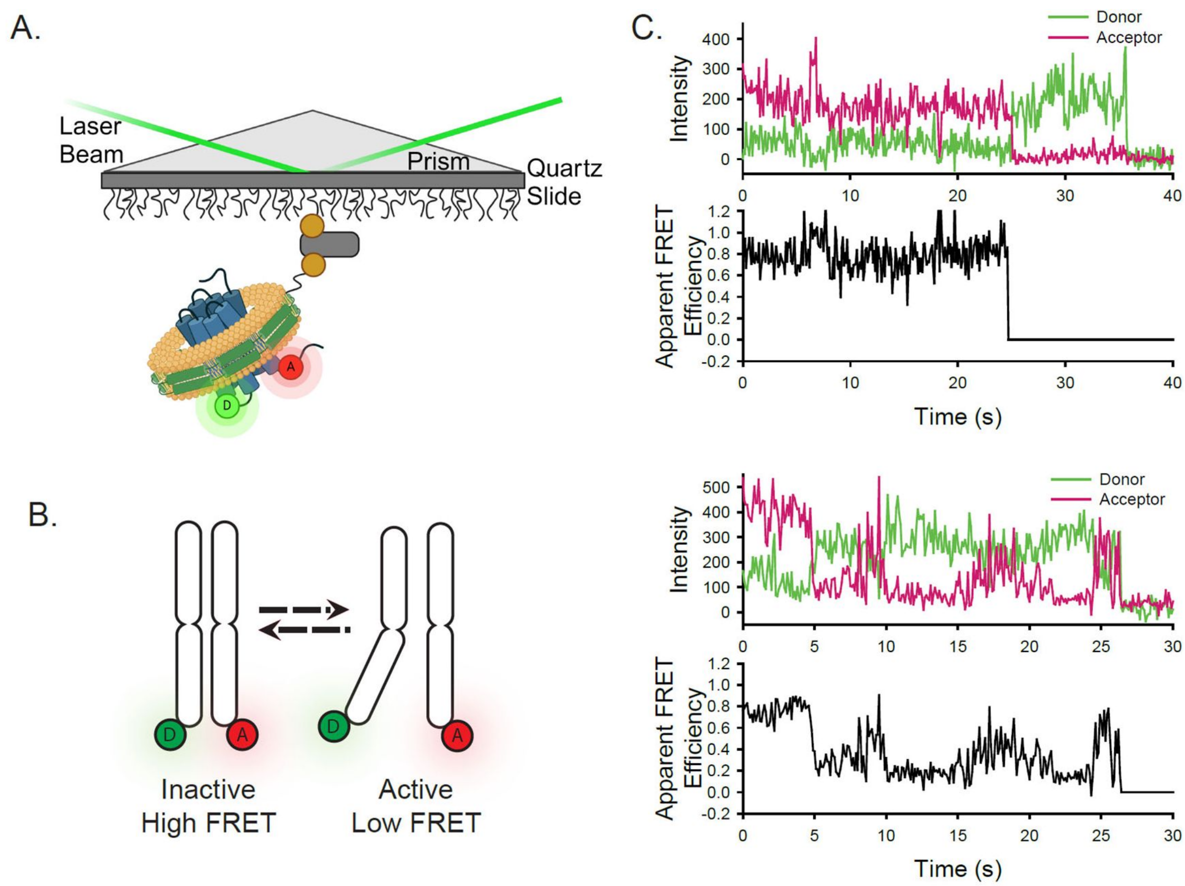

Figure 1.

Experimental design of the smFRET system. A) A single receptor molecule (blue) was labeled with donor (D) and acceptor (A) fluorophores, inserted into a phospholipid (yellow) nanodisc (green), and immobilized on a quartz slide via biotin (tan circle)-neutravidin (grey rectangle) attachment. A prism facilitates total internal reflection of the excitation laser to excite only donor fluorophores close to the surface. B) Cartoon depicting inactive (left) and active (right) receptor conformations. C) Two representative single-molecule time traces for apo-ACKR3. In both examples, the donor (green) and acceptor (red) intensities are shown in the top panel and the corresponding apparent FRET efficiency (black) is shown in the bottom panel.

transitions between inactive and active receptor conformations and would give rise to different donor-acceptor distances and apparent FRET efficiencies that could be resolved by smFRET measurements (shown schematically in **Fig. 1B**). These positions are similar to those used effectively for β_2AR (38), A_{2A}R, (40), and the μ-opioid receptor (47) for monitoring their conformational dynamics by smFRET. Importantly, neither wild-type (WT) CXCR4 nor ACKR3 exhibited significant labeling, obviating the need to remove any of the native cysteine residues in either receptor. Moreover, the double cysteine receptor mutants retained CXCL12-promoted β-arrestin2 recruitment (Figs. S1C & D). Only the E_{max} for arrestin recruitment to CXCL12-stimulated ACKR3 was significantly altered by the mutations, while all other pharmacological parameters were the same as for WT receptors.

ACKR3 exhibits greater conformational dynamics than CXCR4

Receptor-nanodisc complexes were imaged on the slide surface using smFRET microscopy by exciting the A555 donor with a green (532 nm) laser and monitoring the resulting emission from both A555 and the Cy5 acceptor over time on separate segments of a CCD camera. Several hundred individual receptor-nanodisc complexes were typically observed in the field of view. Individual receptor molecules labeled with a single A555 donor and a single Cy5 acceptor were identified by single-step photobleaching transitions (abrupt loss of acceptor fluorescence signal or simultaneous loss of fluorescence signal in both channels), as shown for two representative ACKR3-nanodisc complexes in **Fig. 1C**. Additionally, anti-correlated changes in donor and acceptor emission prior to photobleaching confirmed that FRET occurred between A555 and Cy5. Additional examples of single-molecule traces for ACKR3 and CXCR4 are shown in Fig. S2. Dwell times spent in one FRET level prior to switching to another level are generally in the seconds range (**Figs. 1C** & S2).

The apparent FRET efficiencies from many individual receptor-nanodisc complexes, recorded in the presence or absence of different ligands, were globally analyzed using Hidden Markov Models (HMMs) assuming the presence of two, three, four, or five FRET states (Materials and Methods). To evaluate the appropriate level of model complexity, each resulting apparent FRET efficiency distribution was fit with a Gaussian Mixture Model (GMM) and the corresponding Bayesian Information Criterion (BIC) was calculated. The BIC is a statistical measure of the likelihood that the model describes the data, while also penalizing the addition of parameters that could lead to overfitting of noise. Theoretically, this value will be at a minimum for the model with the appropriate number of states. As an additional criterion, we carefully compared the FRET distributions following the initial HMM analysis for the different models and confirmed that peak positions were consistent across the different experimental conditions.

These analyses indicated that three FRET states were sufficient to fit the smFRET data for CXCR4 under all conditions (Figs. S3 & S4). The resulting apparent FRET efficiency distributions are presented in **Figs. 2A** & B. The predominant high-FRET state (E_{app} = 0.85) observed in the apo-state (**Fig. 2A**) suggests that TM4 and 6 are in close physical proximity, which is consistent with the conformation of inactive GPCRs (48, 49). Accordingly, we interpret this FRET state as the inactive conformation of CXCR4 and designate it as R. The minor low-FRET state (E_{app} = 0.19) reflects outward movement of TM6 away from TM4, as expected for an active receptor conformation (designated R*) (46, 50) and consistent with the limited basal activity of CXCR4. Moreover, there was a major population shift from the high-FRET state to the low-FRET state upon addition of $CXCL12_{WT}$ (**Fig. 2B**), consistent with the expected stabilization of active GPCR conformations by agonists (51, 52). To gain insight into the nature of the mid-FRET state (E_{app} = 0.59), we examined the connectivity between all three FRET states. Two-dimensional transition density probability (TDP) plots revealed that the three FRET states were connected in a sequential fashion (**Figs. 2A** & B), indicating that the transitions occurred within the same molecules. Notably, these observations exclude the possibility that the mid-FRET state arises from different local fluorophore environments (hence FRET efficiencies) for the two possible labeling orientations of the introduced cysteines: assuming two receptor conformations, this model would produce four distinct FRET states, but only two cross peaks in the TDP plot. Another significant observation is that direct transitions between R and R* states were rarely observed (**Figs. 2A** & B). Taken together, these results suggest that the mid-FRET state represents an intermediate receptor conformation (designated R') that lies on the pathway between inactive and active conformations. In the apo-state, transitions were mostly observed between states R and R' (**Fig. 2A**), while in the presence of CXCL12$_{WT}$ the most frequent transitions were observed between the R' and R* states (**Fig. 2B**).

In contrast, the apparent FRET efficiency histogram for ACKR3 in the apo state (**Fig. 2C**) was much broader than the corresponding histogram for CXCR4 (**Fig. 2A**), indicating greater conformational heterogeneity. Moreover, the smFRET data for ACKR3 could not be described by three FRET states and the inclusion of a fourth state was necessary (Figs. S5 & S6). The FRET distributions recovered by four-state global analysis are presented in **Figs. 2C** & D. The positions of the high-FRET (E_{app} = 0.85) and low-FRET (E_{app} = 0.11) peaks are similar to the corresponding peaks observed in CXCR4 and are likewise assigned to inactive (R) and active (R*) conformations, respectively. Consistent with these assignments, the R* state increased in population at the expense of the R state in the presence of the chemokine agonists CXCL12$_{WT}$ (**Fig. 2D**) or CXCL11 (Fig. S7B). TDP plots indicated that the four FRET states in ACKR3 were connected in a sequential fashion and reside along the receptor activation pathway (**Figs. 2C** & D). Similar to the arguments presented above for CXCR4, the intermediate FRET states are most likely discrete receptor conformations and not arising from mixed labeling of the two introduced cysteines. Accordingly, the mid-FRET peaks (E_{app} = 0.39 and E_{app} = 0.66) are assigned to two intermediate receptor conformations, designated R*' and R', respectively. Notably, the R*' state was not observed in CXCR4 (**Figs. 2A** & B). The R' state in ACKR3 decreased in population in the presence of chemokine agonists (**Fig. 2D**, Fig. S7B), suggesting that this state represents an inactive intermediate receptor conformation, consistent with its position on the conformational pathway (closest to R). In striking contrast to apo-CXCR4, apo-ACKR3 populated the four conformational states more or less equally, and all possible sequential conformational transitions were observed (**Fig. 2C**). Thus, ACKR3 is intrinsically more conformationally heterogenous and dynamic than CXCR4. In the presence of CXCL12$_{WT}$, ACKR3 showed more frequent R' ↔ R*' and R*' ↔ R* transitions compared with the apo-receptor, accounting for the population shift towards the R* state (**Figs. 2C** & D).

Effect of small-molecule ligands on conformational states of CXCR4 and ACKR3

The small-molecule ligand IT1t is reported to act as an inverse agonist of CXCR4 (53–55). However, the conformational distribution of CXCR4 showed little change to the overall apparent FRET profile, although R' ↔ R* transitions appeared in the TDP plot (**Figs. 3A** & B, Fig. S8). This suggests that the small molecule does not suppress CXCR4 basal signaling by changing the conformational equilibrium. Instead IT1t appears to increase transition probabilities which may impair G protein coupling by CXCR4.

Treatment of ACKR3 with the small-molecule agonist VUF15485 (56) shifted the conformational distribution of the receptor towards the active R* state (Figs. S7C & S8), as expected for an agonist and supporting the assignment of the R* FRET state. In contrast, treatment of ACKR3 with the small-molecule inverse agonist VUF16840 (52) shifted the conformational distribution to the inactive R conformation, with a concomitant decrease of R* (**Fig. 3D**, Fig. S8), consistent with the expected effect of an inverse agonist and with suppression of the basal activity of ACKR3 by this ligand (52). The intermediate R*' population also decreased in the presence of the inverse agonist (**Fig. 3D**, Fig. S8), suggesting that this state represents an active-like receptor conformation, consistent with its placement on the conformational pathway (closest to R*). The suppression of active receptor conformations was also evident in the TDP, which revealed fewer transitions between R*' and R* conformations relative to the apo receptor (**Figs. 3C** & D).

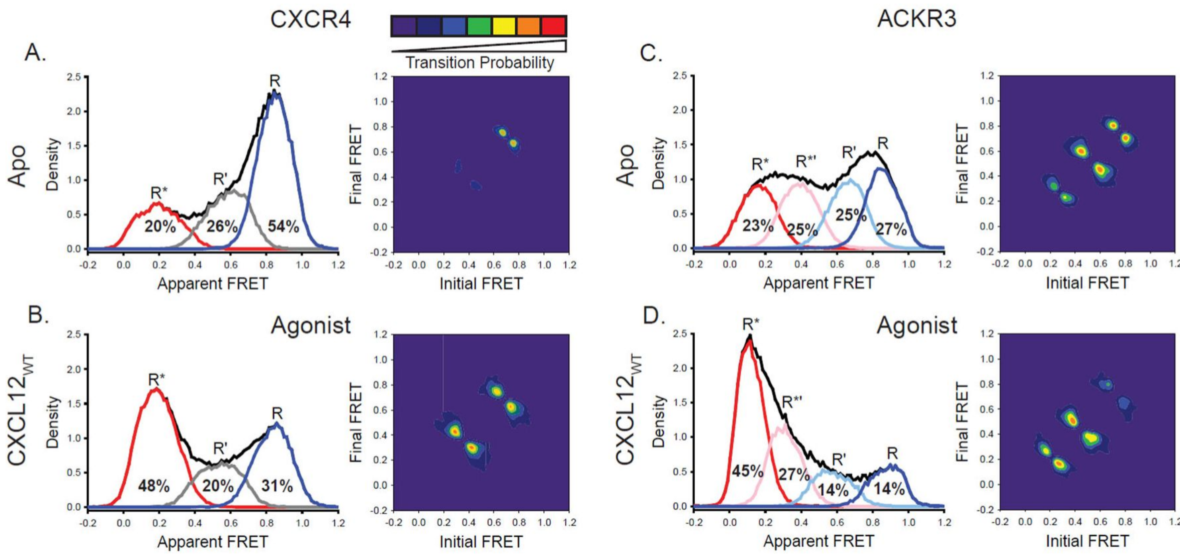

Figure 2.

ACKR3 exhibits greater conformational flexibility compared to CXCR4. A) Apparent FRET efficiency histogram of apo-CXCR4 (left, black trace) resolved into three distinct conformational states: a high-FRET state corresponding to the inactive receptor conformation (R, blue), a low-FRET active receptor conformation (R*, red) and an intermediate conformation (R′, gray). The fractional populations of each state obtained from global analysis are indicated. The receptor is mostly in the inactive conformation. A transition density probability (TDP) plot (right) displays the relative probabilities of transitions from an initial FRET state (x-axis) to a final FRET state (y-axis). For apo-CXCR4, transitions between R and R′ states are observed most frequently. B) Addition of $CXCL12_{WT}$ to CXCR4 shifted the conformational distribution to the low-FRET R* state and resulted in more transitions between all three FRET states. C) The broad apparent FRET efficiency histogram of apo-ACKR3 (left, black trace) is resolved into four distinct conformational states: inactive R (blue), active R* (red), inactive-like R′ (light blue), and active-like R*′ (pink). Little conformational preference is observed among these states. Moreover, all possible sequential state-to-state transitions are observed (right). D) Addition of $CXCL12_{WT}$ to ACKR3 shifted the conformational distribution to the low-FRET R* state, which was also reflected in the transition probabilities. In all cases, data sets represent the analysis of at least three independent experiments.

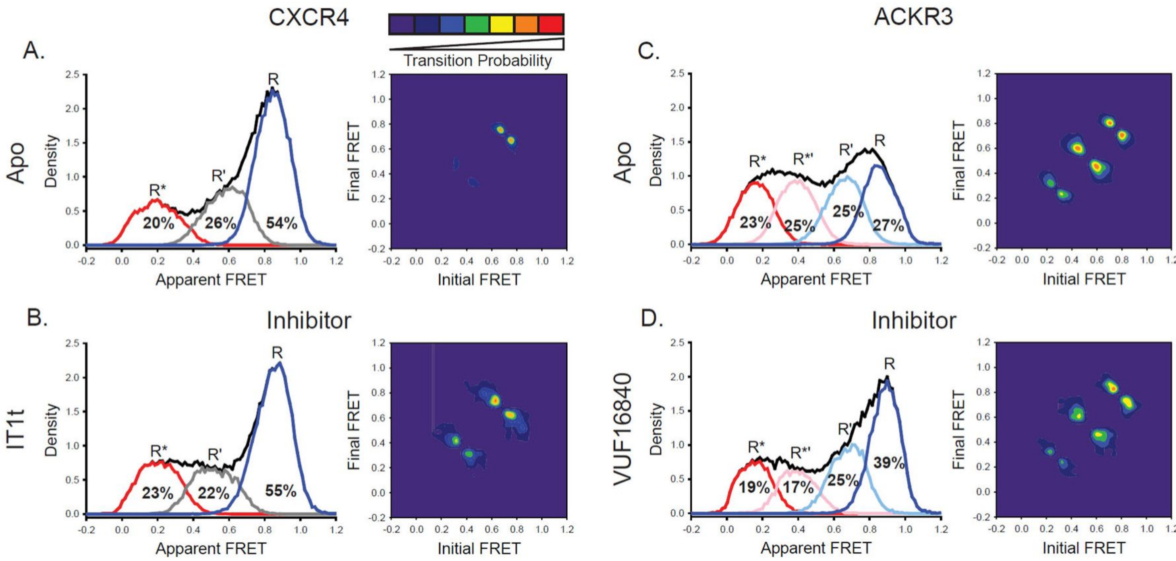

Figure 3.

A small molecule inhibitor shifts the ACKR3 conformational population to the inactive FRET state, while CXCR4 is largely unaffected. A) FRET distributions and TDP of apo-CXCR4 repeated from **Fig. 2A** for comparison. B) Treatment of CXCR4 with the inhibitor IT1t had little impact on the FRET distribution, but increased transition probabilities compared to the apo-receptor. C) FRET distributions and TDP of apo-ACKR3 repeated from **Fig. 2C** for comparison. D) Treatment of ACKR3 with VUF16480, an inverse agonist, shifted the conformational distribution and TDP to the high-FRET inactive R conformation. Data sets represent the analysis of at least three independent experiments. The overall apparent FRET efficiency envelopes for the samples are represented by the black traces.

CXCL12 N-terminal mutants promote active receptor conformations despite their contrasting pharmacological effects on CXCR4 and ACKR3

CXCR4 is sensitive to N-terminal mutations of CXCL12 while ACKR3 is relatively insensitive. For example, the variant $CXCL12_{P2G}$, containing a proline to glycine mutation in the second position, and $CXCL12_{LRHQ}$, where the first three residues of $CXCL12_{WT}$ are replaced with the four-residue motif LRHQ starting with L0, are antagonists of CXCR4 but agonists of ACKR3 (19, 20). To gain further insight into the ligand-dependent responses of CXCR4 and ACKR3, we examined how these mutant chemokines influence the conformational states and dynamics of both receptors.

Surprisingly, $CXCL12_{P2G}$ promoted a shift to the active R* conformation of CXCR4 compared to the apo-receptor (R* increased by 16%, **Figs. 4A** & B, Fig. S8), although the shift was less pronounced than observed for $CXCL12_{WT}$ (28% increase in R*, **Fig. 2B**, Fig. S8). The shift is also evident in the TDPs where the state-to-state transitions involving the R* active state were more probable for the $CXCL12_{P2G}$ complex compared with apo-CXCR4 (**Figs. 4A** & B). The presence of $CXCL12_{LRHQ}$ had a more subtle effect on CXCR4: the active R* conformation increased by 9% (**Figs. 4A** & C, Fig. S8). Despite the ability of $CXCL12_{P2G}$ and $CXCL12_{LRHQ}$ to stabilize the active R* conformation of CXCR4, both variants are known to act as antagonists (20). This suggests that the CXCL12 mutants inhibit CXCR4 coupling to G proteins not by suppressing the active receptor population but rather by increasing the dynamics of the receptor state-to-state transitions. Our results suggest that the helical movements considered classic signatures of the active state may not be sufficient for CXCR4 to engage productively with G proteins..

In the case of ACKR3, $CXCL12_{P2G}$ induced a modest increase in the population of the active R* conformation relative to the apo-receptor (R* increased by 5%, **Figs. 4D** & E, Fig. S8), consistent with the ability of this CXCL12 variant to act as an agonist of ACKR3 (20). Additionally, $CXCL12_{P2G}$ also promoted formation of the active-like intermediate R*' conformation (R*' increased by 6%, **Figs. 4D** & E, Fig. S8), suggesting that activation of ACKR3 can be achieved by populating the R*' state, not just the R* state, which is consistent with a flexible, distortion activation mechanism. Together, $CXCL12_{P2G}$ increased active (R*) and active-like (R*') conformations by 11%, somewhat less than observed for $CXCL12_{WT}$ (24%, **Fig. 2C**, Fig. S8). $CXCL12_{LRHQ}$ also promoted the active (R*) and active-like (R*') conformations of ACKR3 (R* + R*' increased by 11%, **Fig. 4F**, Fig. S8). Additionally, the probabilities of R ↔ R' transitions were suppressed relative to apo receptor in the presence of $CXCL12_{P2G}$, while the probabilities for R'* ↔ R* transitions were enhanced in the presence of the $CXCL12_{LRHQ}$ (**Figs. 4E** & F). These differences may be a consequence of the longer residence time of the LRHQ mutant on the receptor (57). The agonism observed for both chemokine variants (19, 20) suggests that both the active R* and active-like R*' conformations of ACKR3 are sufficient for GRK phosphorylation and arrestin recruitment.

ACKR3 constitutive activity is linked to receptor conformational heterogeneity

As noted above, apo-ACKR3 displays little conformational selectivity, with similar occupancies observed for all four FRET states (**Fig. 2C**). We hypothesized that this might be related to the constitutive activity of the receptor (13, 16) and tied to the presence of Tyr at position $257^{6.40}$ (**Fig. 5A**), which is a hydrophobic residue (V, I, or L) in all other chemokine receptors. In many other class A GPCRs, mutating the residue at 6.40 results in constitutive activity (32, 58–60) and previous analysis of rhodopsin suggests that this is a consequence of lowering the energy barrier for transitions between different receptor conformations (61). In our previous work, we showed that mutation of $Y257^{6.40}$ to leucine, the corresponding amino acid in CXCR4, reduces constitutive arrestin recruitment to ACKR3, while preserving the ability of the receptor to be

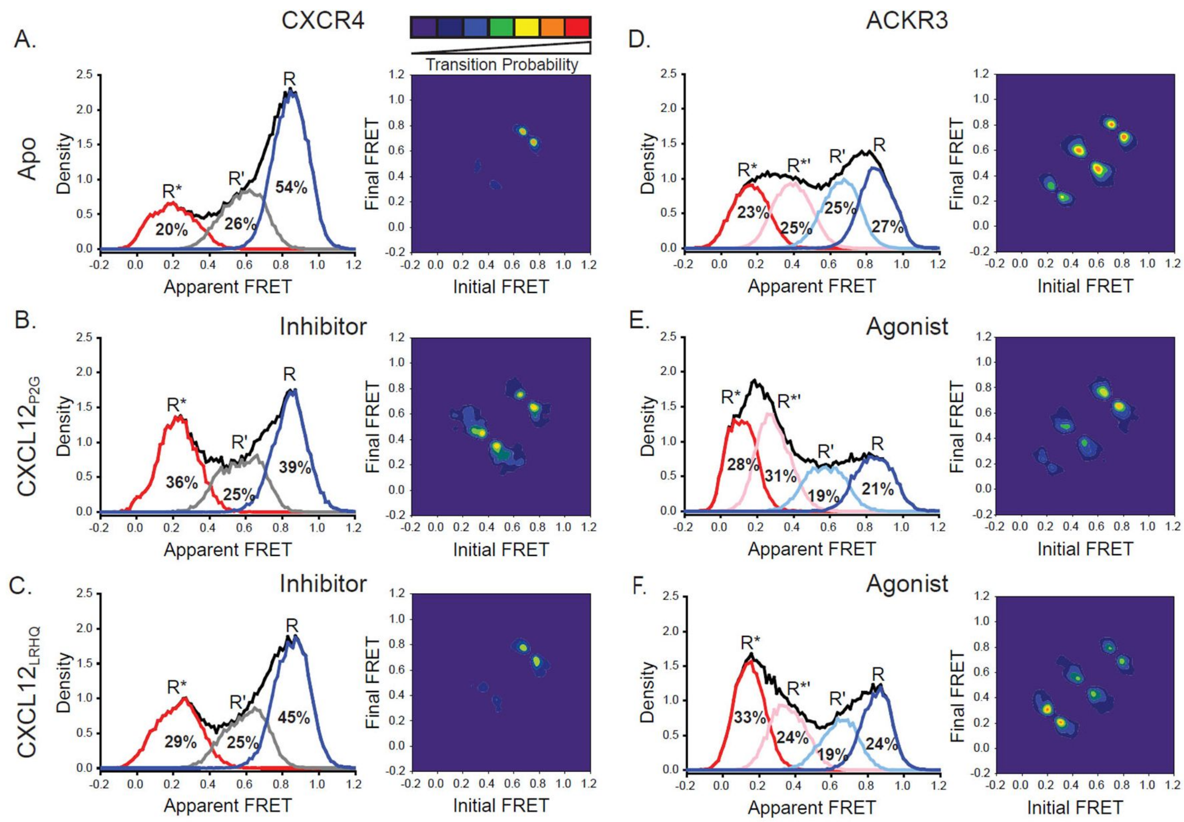

Figure 4.

CXCL12 variants containing mutations to the N-terminus promoted active receptor conformations in both CXCR4 and ACKR3. A) FRET distributions and TDP for apo-CXCR4 repeated from **Fig. 2A** for reference. B) Addition of $CXCL12_{P2G}$ to CXCR4 promoted a shift to the low-FRET active (R*) conformation and an increase in state-to-state transition probabilities. C) $CXCL12_{LRHQ}$ led to a more subtle shift to the R* conformation of CXCR4 without affecting the transition probabilities. D) FRET distributions and TDP for apo-ACKR3 repeated from **Fig. 2C**. E) Treatment of ACKR3 with $CXCL12_{P2G}$ displayed a shift to low-FRET, R*′ and R* states, while reducing the transition probabilities for R′ ↔ R*′ and R*′ ↔ R* transitions relative to the apo-receptor. F) $CXCL12_{LRHQ}$ treatment of ACKR3 shifted the FRET distribution to the low-FRET R* active state and promoted R*′ ↔ R* transitions relative to the apo-receptor. In all cases, the data sets represent the analysis of at least three independent experiments. The overall apparent FRET efficiency envelopes for the samples are represented by the black traces.

activated by CXCL12 (Fig. S9) (16). Adding this single-point mutation to our ACKR3 smFRET construct had a significant impact, converting the broad conformational distribution of the WT apo-receptor to a narrower distribution concentrated in the high-FRET region (**Figs. 5B** & C). Overall, the FRET histogram is similar to what we observe for apo-CXCR4 (**Fig. 2A**), although the active conformation is still split between R* and R*' states (the latter unique to ACKR3) (**Fig. 5C**). State-to-state transitions were also suppressed relative to WT ACKR3 (**Figs. 5B** & C). This result indicates that $Y257^{6.40}$ is a major determinant of the broad conformational heterogeneity of ACKR3.

$CXCL12_{WT}$ promoted active and active-like conformations of $Y257^{6.40}$L ACKR3 (R* + R*' increased by 16%, **Figs. 5C** & E), although the effect was somewhat reduced in comparison to WT ACKR3 (R* + R*' increased by 24%, **Figs. 5B** & D). Consistent with this, the mutant receptor recruits β-arrestin in response to $CXCL12_{WT}$ with an E_{max} value that is only slightly reduced relative to the WT receptor (Fig. S9, ~80% of WT) (16), suggesting again that the population of the R*' intermediate conformational state may contribute to receptor activation.

Discussion

The molecular basis for the atypical pharmacological behavior of ACKR3 is not well understood. Since structures of ACKR3 show intracellular loop disorder, and progressive structural substitutions within the loops fail to promote G protein coupling (16), we recently proposed that the atypical nature of ACKR3 may be related to receptor conformational dynamics (16, 62). Consistent with this hypothesis, in the present study we found that the conformational dynamics of ACKR3 and the canonical GPCR CXCR4 are indeed markedly different. Our smFRET results revealed four distinct conformations of apo-ACKR3 with approximately equal populations: inactive R, active R*, R' (inactive-like) and R*' (active-like) (**Fig. 2C**). The state-to-state TDP plots (**Fig. 2C**) further reinforced the notion that ACKR3 is a flexible receptor that readily exchanges between different conformational states, consistent with our previous structural studies where a flexible intracellular interface in the absence of interaction partners was observed (16). In contrast, apo-CXCR4 primarily populates the inactive R state, with only a single intermediate state (R'), and relatively few transitions between states (**Fig. 2A**). These observations imply that ACKR3 has a relatively flat energy landscape, with similar energy minima for the different conformations, whereas the energy landscape of CXCR4 is more rugged (**Fig. 6**). For both receptors, the energy barriers between states are sufficiently high that transitions occur relatively slowly with seconds long dwell times (**Figs. 1C** and S2).

The relatively flat energy landscape of ACKR3 may account for the propensity of ACKR3 to be activated by a diverse range of ligands. In principle, ligands could promote activation by providing more energetically favorable binding interactions with the receptor in the active R* or R*' conformations relative to the inactive R or R' conformations. Alternatively, ligands could destabilize the inactive R or R' conformations of the receptor, which would also shift the receptor population to the active conformation(s). Our results do not distinguish between these two possibilities. Regardless, the similar free energies of different receptor conformations in ACKR3 (**Fig. 6**) implies that relatively little energy is required to switch the receptor from one state to another. Moreover, activation can be achieved by populating either the R* or R*' state. Thus, independent of specific binding poses and receptor interactions, a variety of different ligands may be able to tip the balance between different receptor conformations.

The TDP plots demonstrate that the R' state in CXCR4 and the R' and R*' states in ACKR3 are intermediate species that lie on the pathway between inactive R and active R* receptor conformations. SmFRET studies of the glucagon receptor (39) and the $A2_{AR}$ (40) (both labeled in TM4 and TM6, as here) also revealed a discrete intermediate receptor conformation. In principle, the intermediates in CXCR4 and ACKR3 could represent partial movements of TM6 from

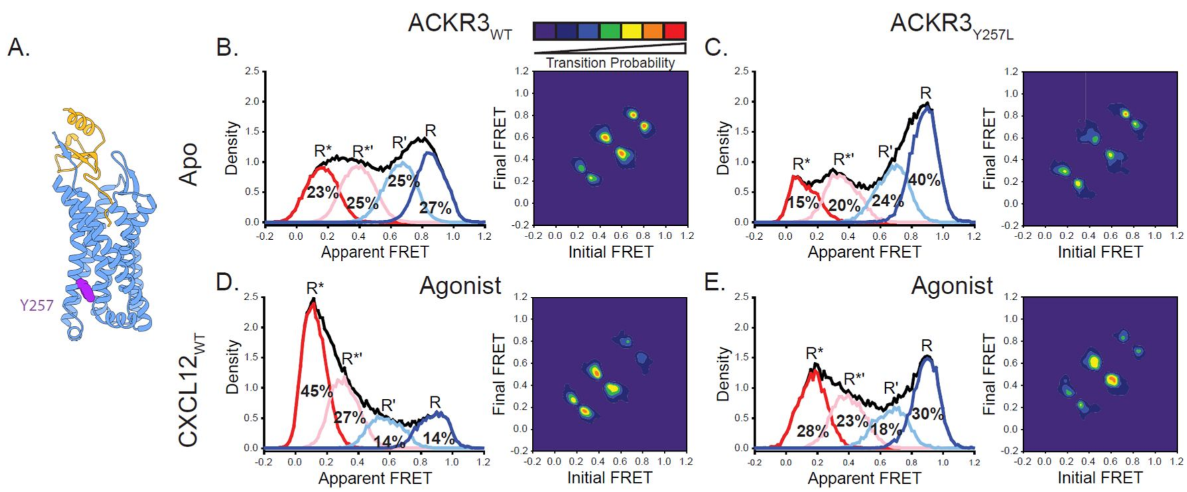

Figure 5.

Replacement of Y257$^{6.40}$ with the corresponding residue in CXCR4 (leucine) reduces conformational heterogeneity of ACKR3. A) Structure of ACKR3 bound with CXCL12$_{WT}$ (PDBID: 7SK3) highlighting the location of Y257$^{6.40}$ (purple) (16). B) Apparent FRET efficiency distributions and TDP of WT ACKR3 in the apo-state, repeated from **Fig. 2C**. C) The mutation Y257$^{6.40}$L shifted the conformational landscape of the apo-receptor to the high-FRET inactive R conformation at the expense of active R* and active-like R*' conformations, and also reduced the probability of state-to-state transitions. D) Apparent FRET efficiency distributions and TDP of WT ACKR3 treated with CXCL12$_{WT}$, repeated from **Fig. 2D**. E) Treatment of Y257$^{6.40}$L ACKR3 with CXCL12$_{WT}$ promoted more low-FRET active R* and active-like R*' states. Data sets represent the analysis of at least three independent experiments. The overall apparent FRET efficiency envelopes for the samples are represented by the black traces.

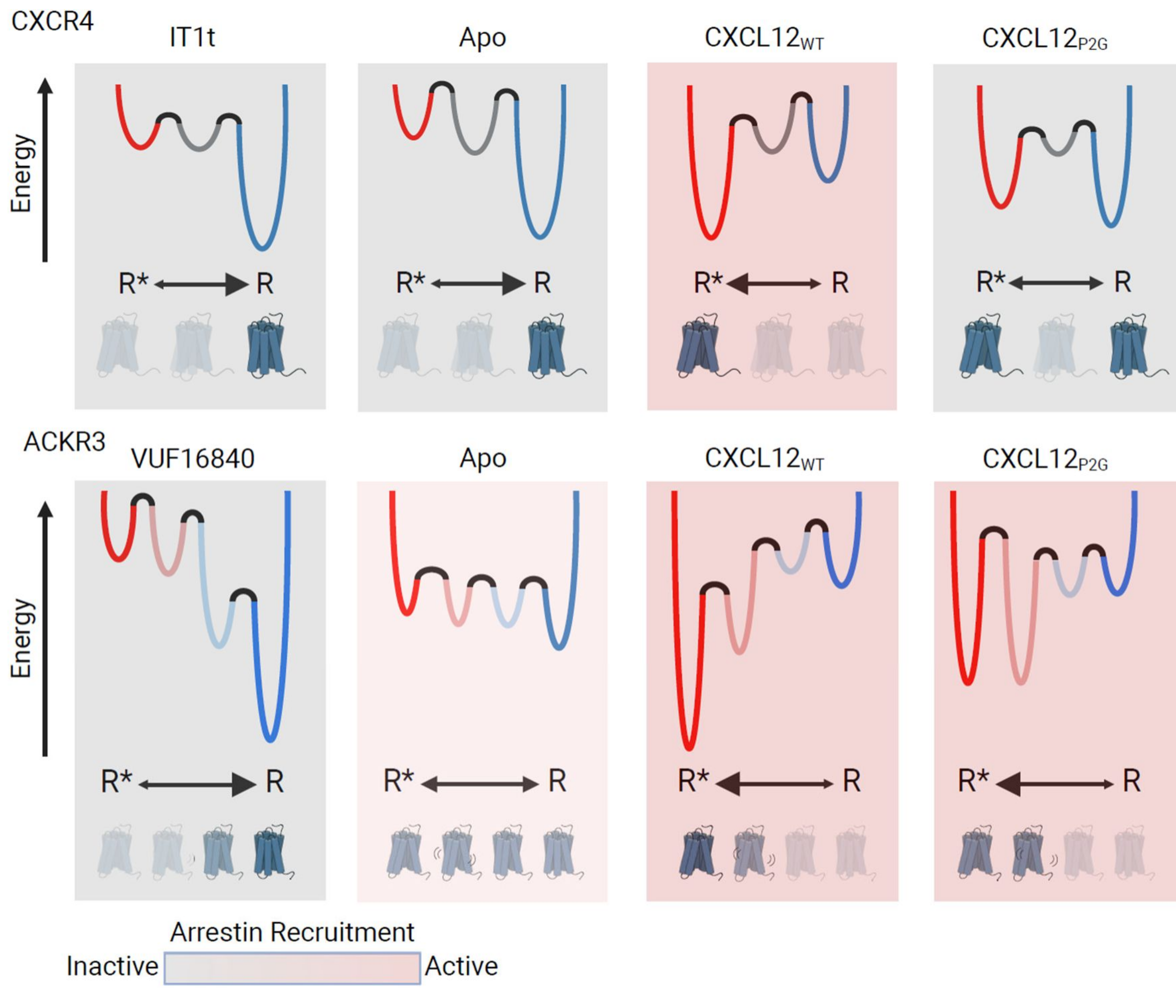

Figure 6.

Schematic illustration of the conformational energy landscapes for CXCR4 and ACKR3, highlighting the differences in the responsiveness of the two receptors to ligands. CXCR4 populates three distinct conformations, shown here as wells on the energy landscape. Apo-CXCR4 is predominantly in the inactive R state. The receptor is converted incompletely to R* with $CXCL12_{WT}$ treatment, while the small molecule inhibitor IT1t has little impact on the conformational distribution. Though $CXCL12_{P2G}$ is an antagonist of CXCR4, the ligand promoted a detectable shift to the active R* state, suggesting TM6 movement is not sufficient for CXCR4 activation. In contrast, ACKR3 populates four distinct conformations and shows little preference among them in the apo-form. The inverse agonist, VUF16840, shifts the population to the inactive R conformation, while the agonists $CXCL12_{WT}$ and $CXCL12_{P2G}$ promote the R* and R*' populations of ACKR3. Despite stabilizing different levels of the active R* state and active-like intermediate R*' state, both $CXCL12_{WT}$ and $CXCL12_{P2G}$ are agonists of ACKR3. The flexibility of ACKR3 may contribute to the ligand-promiscuity of this atypical receptor. Figure created with biorender.com the inactive to active conformation or more subtle conformational changes altering the photophysical characteristics of the probes without drastically altering the donor-acceptor distance. Either possibility leads to detectable changes in apparent FRET efficiency and reflect discrete conformational steps on the activation pathway; however, it is not possible to resolve specific structural changes from the data. Interestingly, the R*' intermediate appears to be unique to ACKR3. Moreover, this state responds to agonist and inverse agonist ligands in the same manner as the active R* state: agonists increase the population of the R*' state, while an inverse agonist decreases the population. In contrast, the single intermediate state in CXCR4 (R') is not responsive or only weakly responsive to ligands. The presence of both active (R*) and active-like (R*') conformations in ACKR3 may be responsible for its activation prone nature and ligand promiscuity.

Structural features that make ACKR3 conformationally flexible include Y257$^{6.40}$, which we previously showed contributes to the constitutive activity of ACKR3 (16). Similar to the constitutively active M257$^{6.40}$Y mutant of rhodopsin, Y$^{6.40}$ in ACKR3 stacks against Y$^{5.58}$ and Y$^{7.53}$ and stabilizes the active-like conformation (63). Additionally, the broad conformational distribution and high probability of state-to-state transitions of ACKR3 parallel the dramatically lower energy barrier of the M257$^{6.40}$Y rhodopsin mutant relative to WT rhodopsin (61). By contrast, mutation of Y257$^{6.40}$ to leucine, the corresponding amino acid in CXCR4, promoted the dominance of an inactive high-FRET receptor conformation similar to CXCR4 (**Figs. 2A** & 5C); it also reduced the transition probabilities (**Figs. 5B** & C) and the ability of ACKR3 to constitutively recruit arrestin (16).

In addition to the contribution of Y257$^{6.40}$ to the dynamic behavior of ACKR3, a disulfide bond between C34 in the receptor N-terminus and C287 in extracellular loop 3 (ECL3), observed in all other reported chemokine receptor-chemokine structures (64–66) is conspicuously missing in cryo-EM structures of ACKR3 with chemokine and small molecule agonists (16). Furthermore, ACKR3 remains functional in the absence of these cysteines (67) unlike other chemokine receptors (68, 69). The disulfide may constrain the relative positions of TM1 and TM6/7 and the opening of the orthosteric pocket; therefore, its absence may confer ACKR3 with greater conformational flexibility than other chemokine receptors, consistent with our observations. It may also allow ACKR3 to be activated by diverse ligands.

Our results provide insights into the linkage between receptor conformation and ligand pharmacology in CXCR4. The chemokine variants CXCL12$_{P2G}$ and CXCL12$_{LRHQ}$ are reported to act as antagonists of CXCR4 (19, 20), and the small molecule IT1t acts as an inverse agonist (54–56). Surprisingly, none of these ligands inhibit formation of the active R* conformation of CXCR4. In fact, the chemokine variants both stabilize and increase this state to some degree, although less effectively than CXCL12$_{WT}$. Thus, the antagonism and inverse agonism of these ligands does not appear to be linked exclusively to receptor conformation, suggesting that the ligands inhibit coupling of G proteins to CXCR4 or disrupt the ligand-receptor-G protein interaction network required for signaling (Fig. S10) (21, 23). Interestingly, these ligands also increase the probabilities of state-to-state transitions (**Figs. 3B** & **4B**), suggesting that enhanced conformational exchange prevents the receptor from productively engaging G proteins. Similarly, ACKR3 is naturally dynamic and lacks G protein coupling, suggesting a common mechanism of G protein antagonism. Future smFRET studies performed in the presence of G proteins should be informative in this regard.

An unusual aspect of ACKR3 behavior is the failure to activate G proteins. Instead, ACKR3 recruits arrestins in response to phosphorylation of its C-terminal tail. Why ACKR3 does not couple to G proteins, at least in most cells, is unclear and not readily explained by differences in primary sequence, since insertion of a DRY box motif and substitution of all ICLs from a canonical GPCR failed to confer G protein activity (16). It is possible that the active receptor conformation clashes sterically with the G protein as suggested by docking of G proteins to structures of ACKR3

(16). Alternatively, the receptor dynamics and conformational transitions revealed here may prevent formation of productive contacts between ACKR3 and G protein that are required for coupling, even though G proteins appear to constitutively associate with the receptor (13, 16, 70). An important caveat is that our study does not report on the dynamics of the other TM helices and H8, some of which are known to participate in arrestin interactions (31, 32). Lack of a well-organized intracellular pocket due to frequent conformational transitions may also explain why the fingerloop of arrestin is not observed to interact with the pocket, in contrast with other GPCRs (71, 72), but instead inserts into membranes/micelles adjacent to the receptor (62). Nevertheless, arrestins are still recruited to CXCL12-stimulated ACKR3 due to GRK phosphorylation of the receptor C-terminal tail (15, 73). Since GRKs also interact with the cytoplasmic pocket to facilitate phosphorylation, it remains to be determined how dynamics might decouple G protein activation and arrestin binding to the receptor cytoplasmic pocket (62) while supporting pocket-mediated GRK activity. However, given the fleeting interaction between GRKs and GPCRs (74), rapid state sampling by ACKR3 may not necessarily be detrimental to GRK engagement and phosphorylation. Furthermore, conformational intermediates in addition to the fully active receptor have been shown to be targets for GRK phosphorylation, such as the early photoactivated rhodopsin metarhodopsin I (75). A more constrained system may be necessary to promote productive interactions between ACKR3 and G proteins. Along these lines, a local increase of membrane pressure in certain cell environments could explain the apparent ability of ACKR3 to activate G proteins in astrocytes and glioma cells (12, 13). The atypical activation of ACKR3 does not appear to be dependent on any singular receptor feature and is likely a combination of several factors.

The pharmacological behavior of ACKR3 resembles the human cytomegalovirus chemokine receptor US28, which also recognizes diverse chemokines, constitutively internalizes, and displays multiple functional conformations (76). Similar to ACKR3, US28 appears to be activated by distortion of the orthosteric binding pocket rather than by specific side chain contacts between receptor and ligand, which is supported by multiple active conformations observed for both apo-US28 and US28 with different agonists (76–79). Whether US28 also has a relatively flat energy landscape like ACKR3 remains to be seen. The conformational dynamics and activation mechanisms revealed here for ACKR3 may also be operative in other chemokine receptors that respond to multiple ligands and have considerable constitutive activity, such as CCR1, CCR2, and CCR3 (64, 80). Finally, the ability of ACKR3 to be activated by populating more than one conformational state may explain why antagonizing the receptor by targeting the orthosteric binding pocket has proven to be challenging; in contrast, the specific requirements for CXCR4 agonism has permitted the development of many orthosteric antagonists but few agonists (81). Drug discovery efforts aimed at inhibiting ACKR3 may therefore require allosteric strategies.

Materials and methods

Unless otherwise stated all chemicals and reagents were purchased from SigmaAldrich or Fisher Scientific. Methoxy e-Coelenterazine (Prolume Purple) was purchased from Nanolight Technologies (Prolume LTD).

Cloning

Human ACKR3 (residues 2-362) preceded by an N-terminal HA signal sequence and followed by C-terminal 10His and FLAG tags was cloned into the pFasBac vector for purification from *Sf9* cells. Human CXCR4 (residues 2-352) with an N-terminal FLAG tag and C-terminal 10His was inserted into pFasBac for purification. For cell-based assays, ACKR3 (residues 2-362) or CXCR4 (residues 2-352) were inserted into pcDNA3.1 expression vector with an N-terminal FLAG tag and followed with a C-terminal *Renilla* luciferase II (ACKR3_rlucII and CXCR4_rlucII). Site-directed mutagenesis was performed by overlap extension and confirmed by Sanger sequencing. No native cysteines were substituted for either CXCR4 or ACKR3.

Arrestin recruitment by BRET

Arrestin recruitment to ACKR3 and CXCR4 was detected using a BRET2 assay as previously described (15, 57). Briefly, HEK293T cells (ATCC) were plated at 750k/well in a 6-well dish in Dulbecco's modified eagle media (DMEM) with 10% fetal bovine serum (FBS) and transfected 24 hrs later with 50 ng receptor_rlucII DNA, 1 µg GFP10_β-arrestin2 (a kind gift from N. Heveker, Université de Montréal, Canada), and 1.4 µg empty pcDNA3.1 vector using TransIT-LT1 transfection system (MirusBio) and expressed for 40 hrs. The cells were then washed with PBS (137 mM NaCl, 2.7 mM KCl, 10 mM Na_2HPO_4, 1.8 mM KH_2PO_4, pH 7.4) and mechanically lifted in Tyrode's buffer (25 mM HEPES, 140 mM NaCl, 1 mM $CaCl_2$, 2.7 mM KCl, 12 mM $NaHCO_3$, 5.6 mM Glucose, 0.5 mM $MgCl_2$, 0.37 mM NaH_2PO_4, pH 7.5). 100k cells were plated per 96-well white BRET plate (BD Fisher) and reattached for 45 min at 37 °C. GFP expression was checked using a SpectraMax M5 plate fluorometer (Molecular Devices) with 485 nm excitation, 538 nm emission, and 530 nm cutoff. 5 µM Prolume Purple substrate was subsequently added and total luminescence detected using a TECAN Spark Luminometer (TECAN Life Sciences) at 37 °C. CXCL12 was then added to each well at the indicated final concentrations and BRET was read using default BRET2 settings (blue emission 360-440 nm, red emission 505-575 nm) and an integration time of 0.5 sec. Experiments were baseline matched and normalized to the E_{max} of WT receptor. The reported data is the average of three independent experiments performed in duplicate. Points were fit to a sigmoidal dose-response model using SigmaPlot 11.0 (Systat Software, Inc).

Receptor purification, labeling, and nanodisc reconstitution

M159$^{4.40}$C/Q245$^{6.28}$C ACKR3 (WT and Y257$^{6.40}$L) and L150$^{4.40}$C/Q233$^{6.29}$C CXCR4 were purified from *Sf9* cells (Expression Systems) as previously described (16). Briefly, *Sf9* cells were infected with baculovirus (prepared using Bac-to-Bac Baculovirus Expression System, Invitrogen) containing either the mutant ACKR3 or CXCR4. Cells were harvested after 48 hrs and membranes dounce homogenized in hypotonic buffer (10 mM HEPES pH 7.5, 10 mM $MgCl_2$, 20 mM KCl) followed three more times with hypotonic buffer with 1 M NaCl. The membranes were spun down at 50k x g for 30 min and resuspended between each round of douncing. After the final round, membranes were incubated with 100 µM CCX662 (Chemocentryx Inc.) for ACKR3 or 100 µM IT1t for CXCR4 and solubilized in 50 mM HEPES pH 7.5, 400 mM NaCl, 0.75/0.15% dodecyl maltoside/cholesteryl hemisuccinate (DDM/CHS) with a protease inhibitor tablet (Roche) for 4 hrs. Insoluble material was then removed by centrifugation at 50k x g for 30 min and Talon resin (Clontech) with 20 mM imidazole overnight binding at 4 °C. The resin was then transferred to a column and washed with WB1 (50 mM HEPES pH 7.5, 400 mM NaCl, 0.1/0.02% DDM/CHS, 10% glycerol, 20 mM imidazole) followed by WB2 (WB1 with 0.025/0.005% DDM/CHS) and finally eluted with WB2 with 250 mM imidazole. The imidazole was removed by desalting column (PD MiniTrap G-25, GE Healthcare). Final protein concentration was determined by A_{280} using an extinction coefficient of 75000 $M^{-1}cm^{-1}$ (ACKR3) and 58850 $M^{-1}cm^{-1}$ (CXCR4). Samples were snap frozen in liquid nitrogen, and stored at -80 °C until use.

When ready to prepare samples for smFRET measurements, two nanomoles of receptor was thawed and incubated with fourteen nanomoles of Alexa Fluor 555 C2 maleimide (A555) and Cy5 maleimide overnight at 4 °C with rotation. The next morning, free label was removed by dilution using a 100k Da cut-off spin concentrator (Amicon) and the sample concentrated to ~100 µl. Label incorporation was evaluated by measuring the absorbance at A_{280} (ε_{280} = 75000 $M^{-1}cm^{-1}$ for ACKR3 and 58850 $M^{-1}cm^{-1}$ for CXCR4), A_{555} (ε_{555} = 150000 $M^{-1}cm^{-1}$), and A_{645} (ε_{645} = 250000 $M^{-1}cm^{-1}$) to detect the concentrations of the labeled receptor, A555, and Cy5, respectively. The contribution of the fluorophores to A_{280} was removed before determining labeled receptor concentration. The entire sample was used for nanodisc reconstitution.

Labeled receptors were reconstituted into biotinylated MSP1E3D1 nanodiscs as previously described (16). Briefly, 1-palmitoyl-2-oleoyl-*sn*-glycero-3-phosphocholine (POPC, Avanti) and 1-palmitoyl-2-oleoyl-*sn*-glycero-3-phospho-(1'-rac-glycerol) (POPG, Avanti) were prepared in a 3:2 POPC:POPG ratio and solubilized in ND buffer (25 mM HEPES pH 7.5, 150 mM NaCl, 180 mM cholate). MSP1E3D1 was expressed and purified as previously described (82) and biotinylated with EZ-Link NHS-polyethylene glycol 4 (PEG4)-Biotin (Thermo Fisher) per manufacturer instructions. The receptors, MSP1E3D1, and lipids were combined at a molar ratio of 0.1:1:110 for ACKR3:MSP:lipids respectively. Additional ND buffer was added to keep the final cholate concentration >20 mM. After 30 min at 4 °C, 200 mg of Biobeads (Bio-Rad) were added and incubated for 3-6 hrs. The sample was then loaded on a Superdex 200 10/300 GL column equilibrated with 25 mM HEPES pH 7.5, 150 mM NaCl and fractions containing nanodisc complexes were combined. 200 µl Talon resin was added with 20 mM imidazole (final concentration) and the samples were incubated for 16 hrs at 4 °C. The resin was then transferred to a Micro Bio-Spin Column (Bio-Rad) and washed with 25 mM HEPES pH 7.5, 150 mM NaCl, 20 mM imidazole and eluted with 25 mM HEPES pH 7.5, 150 mM NaCl, 250 mM imidazole. Imidazole was removed by buffer-exchange using 100k Da spin concentrators and 25 mM HEPES pH 7.5, 150 mM NaCl. Sample were concentrated to ~1 µM and stored at 4 °C until use.

CXCL12 purification from *E. Coli*

CXCL12 was expressed and purified as previously described (15, 16). Briefly, the chemokines were expressed by IPTG induction in BL21(DE3)pLysS cells. The cells were collected by centrifugation, resuspended in 50 mM tris pH 7.5, 150 mM NaCl and lysed by sonication. Inclusion bodies were then collected by centrifugation, resuspended in equilibration buffer (50 mM tris, 6 M guanidine-HCl pH 8.0), sonicated to release the chemokines and the samples centrifuged again to pellet insoluble material. The supernatant was then passed over a Ni-nitrilotriacetic acid (NTA) column equilibrated with equilibration buffer to bind the His-tagged chemokines. The column was washed with wash buffer (50 mM MES pH 6.0, 6 M guanidine-HCl) and eluted with 50 mM acetate pH 4.0, 6 M guanidine-HCl. The chemokine-containing elutions were pooled and dithiothreitol (DTT) added to a final concentration of 4 mM. After incubating 10 min, the solution was added dropwise into refolding buffer (50 mM tris pH 7.5, 500 mM arginine-HCl, 1 mM EDTA, 1 mM oxidized glutathione) and incubated at room temperature for 4 hrs before dialyzing against 20 mM tris pH 8.0, 50 mM NaCl. To remove the N-terminal purification tag, enterokinase was added and the sample incubated at 37 °C for 5 days. Uncleaved chemokine and free tags were removed by reverse Ni-NTA and eluted with wash buffer. Finally, the sample was purified on a reverse-phase C18 column equilibrated with 75% buffer A (0.1% trifluoroacetic acid (TFA)) and 25% buffer B (0.1% TFA, 90% acetonitrile) and eluted by a linear gradient of buffer B. The pure protein was lyophilized and stored at -80 °C until use.

smFRET microscopy

smFRET experiments were performed on a custom built prism-based TIRF microscope as previously described (83). Briefly, a flow cell was assembled on a quartz slide passivated with polyethylene glycol (PEG) and a small fraction of biotinylated PEG, after which neutravidin was introduced (84). Labeled ACKR3 or CXCR4 in biotinylated nanodiscs were diluted into trolox buffer (25 mM HEPES pH 7.5, 150 mM NaCl, 1 mM propyl gallate, 5 mM trolox), flowed into the sample chamber, and incubated for 5 min at room temperature. The samples were then washed twice with imaging buffer (trolox buffer with 2 mM protocatechuic acid, 50 nM protocatechuate-3,4-dioxygenase). Movies were collected with 100 ms integration time using a custom single-molecule data acquisition program to control the CCD camera (Andor). Single-molecule donor and acceptor emission traces were extracted from the recordings using custom IDL (Interactive Data Language) scripts. The software packages used to control the CCD camera and extract time trajectories were provided by Dr. Taekjip Ha. In all cases, five initial apo-receptor movies were recorded at different locations on the slide and then ligand was flowed into the cell by two washes with chemokine or small molecule in imaging buffer at final concentrations of 500 nM for chemokines ($CXCL12_{WT}$, $CXCL12_{P2G}$, $CXCL12_{LRHQ}$, CXCL11) and 1 μM for the small molecules (IT1t, VUF16840, VUF15485). Ten more movies were collected for each condition at different locations on the slide. The data presented are a composite of at least three individual slides and treatments.

FRET trajectories were generated and analyzed using custom software written in-house (*https://github.com/rpauszek/smtirf*). Donor and acceptor traces for each molecule were corrected for donor bleed through and background signal and apparent FRET efficiencies were calculated as $E_{app} = I_A/(I_A+I_D)$, where E_{app} is the apparent FRET efficiency at each time point and I_D and I_A are the corresponding donor and acceptor fluorophore intensities, respectively. Traces were screened manually for single donor and acceptor bleach steps and anti-correlated behavior between fluorophores to confirm the presence of single receptors within the identified particles and single donor and acceptor labeling. All traces for a particular protein/ligand combination were analyzed globally by a single Hidden Markov Model assuming two, three, four, or five states and shared variance as previously described (83). Briefly, each model was trained on all selected trajectories for a given sample simultaneously using an expectation-maximization method (85). Once a model was trained, the Viterbi algorithm (85) was used to determine the most likely hidden path for each trajectory. This labeled state path was then used to aggregate all data points belonging to a particular state in order to compile composite histograms of apparent FRET efficiency, using a Kernel Density estimation algorithm (Python package Scikit-learn, version 1.2.2) with a Gaussian kernel and a bandwidth of 0.04. The relative populations of distinct FRET states were directly obtained during compilation of the corresponding histograms. The resulting apparent FRET efficiency histograms were fit with a Gaussian Mixture Model (Scikit-learn) and the corresponding Bayesian Information Criterion (BIC) was calculated as described (86). Transition density probability plots revealing the connectivity among individual FRET states were calculated as described (87).

Data availability

All relevant data supporting this study are included in the Article or Supplementary files and are available from the authors upon request.

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Additional information

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Additional files

Supplemental Information

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Editors

Joint Public Review:

Summary

This manuscript uses single-molecule fluorescence resonance energy transfer (smFRET) to identify differences in the molecular mechanisms of CXCR4 and ACKR3, two 7-transmembrane receptors that both respond to the chemokine CXCL12 but otherwise have very different signaling profiles. CXCR4 is highly selective for CXCL12 and activates heterotrimeric G proteins. In contrast, ACKR3 is quite promiscuous and does not couple to G proteins, but like most G protein-coupled receptors (GPCRs), it is phosphorylated by GPCR kinases and recruits arrestins. By monitoring FRET between two positions on the intracellular face of the receptor (which highlight the movement of transmembrane helix 6 [TM6], a key hallmark of GPCR activation), the authors show that CXCR4 remains mostly in an inactive-like state until CXCL12 binds and stabilizes a single active-like state. ACKR3 rapidly exchanges among four different conformations even in the absence of ligand, and agonists stabilize multiple activated states.

Strengths

The core method employed in this paper, smFRET, can reveal dynamic aspects of these receptors (the breadth of conformations explored and the rate of exchange among them) that are not evident from static structures or many other biophysical methods. smFRET has not been broadly employed in studies of GPCRs. Therefore, this manuscript makes important conceptual advances in our understanding of how related GPCRs can vary in their conformational dynamics.

Weaknesses

The probes used cannot reveal conformational changes in other positions besides transmembrane helix 6 (TM6). GPCRs are known to exhibit loose allosteric coupling, so the conformational distribution observed at TM6 may not fully reflect the global conformational distribution of receptors. This could mask important differences that determine the ability of intracellular transducers to couple to specific receptor conformations.

While it is clear that CXCR4 and ACKR3 have very different conformational dynamics, the data do not definitely show that this is the main or only mechanism that contributes to their functional differences.

The extent to which conformational heterogeneity is a characteristic feature of ACKRs that contributes to their promiscuity and arrestin bias is unclear. The key residue the authors find promotes ACKR3 conformational heterogeneity is not conserved in most other ACKRs, but alternative mechanisms could generate similar heterogeneity.

An inherent limitation of the approach is that mutagenesis, purification, and labeling of the receptors could affect their conformational distributions. The cysteine mutations in ACKR3 required to site-specifically install fluorophores substantially increase its ligand-induced activity (Fig. S1D). There are no data to confirm that the two receptors retain the same functional profiles observed in cell-based systems following in vitro manipulations (purification, labeling, nanodisc reconstitution).

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Author response:

The following is the authors’ response to the original reviews.

> ***Reviewer #1 (Public Review):***
>
> *Summary:*
>
> *This paper uses single-molecule FRET to investigate the molecular basis for the distinct activation mechanisms between 2 GPCR responding to the chemokine CXCL12 : CXCR4, that couples to G-proteins, and ACKR3, which is G-protein independent and displays a higher basal activity.*
>
> *Strengths:*
>
> *It nicely combines the state-of-the-art techniques used in the studies of the structural dynamics of GPCR. The receptors are produced from eukaryotic cells, mutated, and labeled with single molecule compatible fluorescent dyes. They are reconstituted in nanodiscs, which maintain an environment as close as possible to the cell membrane, and immobilized through the nanodisc MSP protein, to avoid perturbing the receptor's structural dynamics by the use of an antibody for example.*
>
> *The smFRET data are analysed using the HHMI technique, and the number of states to be taken into account is evaluated using a Bayesian Information Criterion, which constitutes the state-of-the-art for this task.*
>
> *The data show convincingly that the activation of the CXCR4 and ACKR3 by an agonist leads to a shift from an ensemble of high FRET states to an ensemble of lower FRET states, consistent with an increase in distance between the TM4 and TM6. The two receptors also appear to explore a different conformational space. A wider distribution of states is observed for ACKR3 as compared to CXCR4, and it shifts in the presence of agonists toward the active states, which correlates well with ACKR3's tendency to be constitutively active. This interpretation is confirmed by the use of the mutation of Y254 to leucine (the corresponding residue in CXCR4), which leads to a conformational distribution that resembles the one observed with CXCR4. It is correlated with a decrease in constitutive activity of ACKR3.*
>
> *Weaknesses:*
>
> *Although the data overall support the claims of the authors, there are however some details in the data analysis and interpretation that should be modified, clarified, or discussed in my opinion*

Concerning the amplitude of the changes in FRET efficiency: the authors do not provide any structural information on the amplitude of the FRET changes that are expected. To me, it looks like a FRET change from ~0.9 to ~0.1 is very important, for a distance change that is expected to be only a few angstroms concerning the movement of the TM6. Can the authors give an explanation for that? How does this FRET change relate to those observed with other GPCRs modified at the same or equivalent positions on TM4 and TM6?

The large FRET change in our system was initially unexpected. However, the reviewer is mistaken that the expected distance change is only a few angstroms. Crystal structures of the homologous beta2 adrenergic receptor (β_2AR) in inactive and active conformations reveal that the cytoplasmic end of TM6 moves outwards by 16 angstroms during activation (Rasmussen et al., 2011, ref 47). Consistent with this, smFRET studies of β_2AR labeled in TM4 and TM6 (as here) showed that the donor-acceptor (D-A) distance was 14 angstroms longer in the active conformation (Gregorio et al., ref 38). Surprisingly, the apparent distance change in our system (calculated for our FRET probes, A555/Cy5, using FPbase.com) is almost 30 angstroms. A possible explanation is that the fluorophore attached to TM6 interacts with lipids within the nanodisc when TM6 moves outwards, which could stretch the fluorophore linker and thereby increase the D-A distance (lipids were absent in the β_2AR study). Such an interaction could also constrain the fluorophore in an unfavorable orientation for energy transfer, also leading to lower than expected FRET efficiencies and inflated distance calculations. Regardless, it is important to emphasize that none of the interpretations or conclusions of our study are based on computed D-A distances. Rather, we resolved different receptor conformations and quantified their relative populations based on the measured FRET efficiency distributions.

Finally, we note that a recent smFRET study of the glucagon receptor (labeled in TM4 and TM6, as here) also revealed a large difference in apparent FRET efficiencies between inactive (E_{app} = 0.83) and active (E_{app} = 0.32) conformations (Kumar et al., ref. 39). Thus, the large change in FRET efficiency observed in our study is not unprecedented.

Concerning the intermediate states: the authors observe several intermediate states.

(1) First I am surprised, looking at the time traces, by the dwell times of the transitions between the states, which often last several seconds. Is such a long transition time compatible with what is known about the kinetic activation of these receptors?

We too were surprised by the apparent kinetics of the receptors in our system. However, it was previously noted that purified systems, including nanodiscs, lead to slower activation times for GPCRs compared to cellular membrane systems (Lohse et al, Curr. Opin. Cell Biology, 27, 8792, 2014). Indeed, slow transitions among different FRET states (dwell times in the seconds range) were also observed in recent smFRET studies of the mu opioid receptor (Zhao et al., 2024, ref. 41) and the glucagon receptor (Kumar et al., 2023, ref. 39). These studies are consistent with the observed time scale of the FRET transitions reported here.

(2) Second is it possible that these "intermediate" states correspond to differences in FRET efficiencies, that arise from different photophysical states of the dyes? Alexa555 and Cy5 are Cyanines, that are known to be very sensitive to their local environment. This could lead to different quantum yields and therefore different FRET efficiencies for a similar distance. In addition, the authors use statistical labeling of two cysteines, and have therefore in their experiment a mixture of receptors where the donor and acceptor are switched, and can therefore experience different environments. The authors do not speculate structurally on what these intermediate states could be, which is appreciated,

> *but I think they should nevertheless discuss the potential issue of fluorophore photophysics effects.*

The reviewer is correct that the intermediate FRET states could, in principle, arise from a conformational change of the receptor that alters the local environment of the donor and/or acceptor fluorophores, rather than a change in donor-acceptor distance. This caveat is now included in the discussion on Pg. 10:

"In principle, the intermediates in CXCR4 and ACKR3 could represent partial movements of TM6 from the inactive to active conformation or more subtle conformational changes altering the photophysical characteristics of the probes without drastically altering the donor-acceptor distance. Either possibility leads to detectable changes in apparent FRET efficiency and reflect discrete conformational steps on the activation pathway; however, it is not possible to resolve specific structural changes from the data."

Regarding the second possibility, it is true that our labeling methodology leads to a statistical mixture of labeled species (D on TM6 and A on TM4, D on TM4 and A on TM6). If the photophysical properties of the fluorophores were markedly different for the two labeling orientations, this would produce two different FRET efficiencies for a given receptor conformation. Assuming two receptor conformations, this scenario would produce four distinct FRET states: E_1 (inactive receptor, labeling configuration 1), E_2 (active receptor, labeling configuration 1), E_3 (inactive receptor, labeling configuration 2) and E_4 (active receptor, labeling configuration 2), with two cross peaks in the TDP plots, corresponding to $E_1 \leftrightarrow E_2$ and $E_3 \leftrightarrow E_4$ transitions. Notably, $E_2 \leftrightarrow E_3$ cross peaks would not be present, since states E_2 and E_3 exist on separate molecules. Instead, we see all states inter-connected sequentially, R ↔ R' ↔ R* in CXCR4 and R ↔ R' ↔ R*' ↔ R* in ACKR3 (Fig. 2), suggesting that the resolved FRET states represent interconnected conformational states.

We added the following text to the Results section on Pg. 6:

"Two-dimensional transition density probability (TDP) plots revealed that the three FRET states were connected in a sequential fashion (Figs. 2A & B), indicating that the transitions occurred within the same molecules. Notably, these observations exclude the possibility that the midFRET state arises from different local fluorophore environments (hence FRET efficiencies) for the two possible labeling orientations of the introduced cysteines: assuming two receptor conformations, this model would produce four distinct FRET states, but only two cross peaks in the TDP plot."

> *(3) It would also have been nice to discuss whether these types of intermediate states have been observed in other studies by smFRET on GPCR labeled at similar positions.*

Intermediate states have also been reported in previous smFRET studies of other GPCRs. For example, in the glucagon receptor (also labeled in TM4 and TM6), a third FRET state (E_{app} = 0.63) was resolved between the inactive (E_{app} = 0.85) and active (E_{app} = 0.32) states (Kumar et al., Ref. 39). Discrete intermediate receptor conformations were also observed in the $A_{2A}R$ labeled in TM4 and TM6 (Fernandes et al., Ref 40). These examples are now cited in the Discussion.

> *On line 239: the authors talk about the R↔R' transitions that are more probable. In fact it is more striking that the R'↔R* transition appears in the plot. This transition is a signature of the behavior observed in the presence of an agonist, although IT1t is supposed to be an inverse agonist. This observation is consistent with the unexpected (for an inverse agonist) shift in the FRET histogram distribution. In fact, it appears that all CXCR4 antagonists or inverse agonists have a similar (although smaller) effect than the agonist. Is this related to the fact that these (antagonist or inverse agonist) ligands lead*

> *to a conformation that is similar to the agonists, but cannot interact with the G-protein ?? Maybe a very interesting experiment would be here to repeat these measurements in the presence of purified G-protein. G-protein has been shown to lead to a shift of the conformational space explored by GPCR toward the active state (using smFRET on class A and class C GPCR). It would be interesting to explore its role on CXCR4 in the presence of these various ligands. Although I am aware that this experiment might go beyond the scope of this study, I think this point should be discussed nevertheless.*

We thank the reviewer for this observation and the possible explanation offered. In response, we have added the following text to the Results section on Pg. 7:

"The small-molecule ligand IT1t is reported to act as an inverse agonist of CXCR4 (54-56). However, the conformational distribution of CXCR4 showed little change to the overall apparent

FRET profile, although R' ↔ R* transitions appeared in the TDP plot (Figs. 3A & B, Fig. S8). This suggests that the small molecule does not suppress CXCR4 basal signaling by changing the conformational equilibrium. Instead IT1t appears to increase transition probabilities which may impair G protein coupling by CXCR4."

We have also added the following text to the Results on Pg. 8:

"Despite the ability of $CXCL12_{P2G}$ and $CXCL12_{LRHQ}$ to stabilize the active R* conformation of CXCR4, both variants are known to act as antagonists (20). This suggests that the CXCL12 mutants inhibit CXCR4 coupling to G proteins not by suppressing the active receptor population but rather by increasing the dynamics of the receptor state transitions. Our results suggest that the helical movements considered classic signatures of the active state may not be sufficient for CXCR4 to engage productively with G proteins."

In addition, we have added the following text to the Discussion on Pg. 11:

"The chemokine variants $CXCL12_{P2G}$ and $CXCL12_{LRHQ}$ are reported to act as antagonists of CXCR4 (19, 20), and the small molecule IT1t acts as an inverse agonist (54-56). Surprisingly, none of these ligands inhibit formation of the active R* conformation of CXCR4. In fact, the chemokine variants both stabilize and increase this state to some degree, although less effectively than $CXCL12_{WT}$. Thus, the antagonism and inverse agonism of these ligands does not appear to be linked exclusively to receptor conformation, suggesting that the ligands inhibit coupling of G proteins to CXCR4 or disrupt the ligand-receptor-G protein interaction network required for signaling (Fig. S10) (21, 23). Interestingly, these ligands also increase the probabilities of state-to-state transitions (Figs. 3B & 4B), suggesting that enhanced conformational exchange prevents the receptor from productively engaging G proteins. Similarly, ACKR3 is naturally dynamic and lacks G protein coupling, suggesting a common mechanism of G protein antagonism."

Finally, we also agree that experiments with G proteins could be informative. In fact, we initiated such experiments during the course of this study. However, it soon became apparent that significant optimization would be required to identify fluorophore labeling positions that report receptor conformation without inhibiting G protein coupling. Accordingly, we decided that G protein experiments would be the subject of future studies.

However, we added the following text to the Discussion on Pg. 12:

"Future smFRET studies performed in the presence of G proteins should be informative in this regard".

> *The authors also mentioned in Figure 6 that the energetic landscape of the receptors is relatively flat ... I do not really agree with this statement. For me, a flat conformational*

> *landscape would be one where the receptors are able to switch very rapidly between the states (typically in the submillisecond timescale, which is the timescale of protein domain dynamics). Here, the authors observed that the transition between states is in the second timescale, which for me implies that the transition barrier between the states is relatively high to preclude the fast transitions.*

We thank the reviewer for the comment. We have modified the description of the energy landscapes of ACKR3 and CXCR4 in the discussion on Pg. 10 as follows:

"These observations imply that ACKR3 has a relatively flat energy landscape, with similar energy minima for the different conformations, whereas the energy landscape of CXCR4 is more rugged (Fig. 6). For both receptors, the energy barriers between states are sufficiently high that transitions occur relatively slowly with seconds long dwell times (Figs. 1C and S2)."

> ***Reviewer #2 (Public Review):***
>
> *Summary:*
>
> *his manuscript uses single-molecule fluorescence resonance energy transfer (smFRET) to identify differences in the molecular mechanisms of CXCR4 and ACKR3, two 7transmembrane receptors that both respond to the chemokine CXCL12 but otherwise have very different signaling profiles. CXCR4 is highly selective for CXCL12 and activates heterotrimeric G proteins. In contrast, ACKR3 is quite promiscuous and does not couple to G proteins, but like most G protein-coupled receptors (GPCRs), it is phosphorylated by GPCR kinases and recruits arrestins. By monitoring FRET between two positions on the intracellular face of the receptor (which highlights the movement of transmembrane helix 6 [TM6], a key hallmark of GPCR activation), the authors show that CXCR4 remains mostly in an inactive-like state until CXCL12 binds and stabilizes a single active-like state. ACKR3 rapidly exchanges among four different conformations even in the absence of ligands, and agonists stabilize multiple activated states.*
>
> *Strengths:*
>
> *The core method employed in this paper, smFRET, can reveal dynamic aspects of these receptors (the breadth of conformations explored and the rate of exchange among them) that are not evident from static structures or many other biophysical methods. smFRET has not been broadly employed in studies of GPCRs. Therefore, this manuscript makes important conceptual advances in our understanding of how related GPCRs can vary in their conformational dynamics.*
>
> *Weaknesses:*
>
> *(1) The cysteine mutations in ACKR3 required to site-specifically install fluorophores substantially increase its basal and ligand-induced activity. If, as the authors posit, basal activity correlates with conformational heterogeneity, the smFRET data could greatly overestimate the conformational heterogeneity of ACKR3.*

The change in basal ACKR3 activity with the Cys introductions are modest in comparison and insignificantly different as determined by extra-sum-of-squares F test (P=0.14).

> *(2) The probes used cannot reveal conformational changes in other positions besides TM6. GPCRs are known to exhibit loose allosteric coupling, so the conformational distribution observed at TM6 may not fully reflect the global conformational distribution of receptors. This could mask important differences that determine the ability of intracellular transducers to couple to specific receptor conformations.*

We agree that the overall conformational landscape of the receptors has not been investigated and we have added this caveat to the discussion on Pg. 12.

"An important caveat is that our study does not report on the dynamics of the other TM helices and H8, some of which are known to participate in arrestin interactions."

> *(3) While it is clear that CXCR4 and ACKR3 have very different conformational dynamics, the data do not definitively show that this is the main or only mechanism that contributes to their functional differences. There is little discussion of alternative potential mechanisms.*

The main functional difference between CXCR4 and ACRK3 is their effector coupling: CXCR4 couples to G proteins, whereas ACKR3 only couples to arrestins (following phosphorylation of the C-terminal tail by GRKs). As currently noted in the discussion, ACKR3 has many features that may contribute to its lack of G protein coupling, including lack of a well-ordered intracellular pocket due to conformational dynamics, lack of an N-term-ECL3 disulfide, different chemokine binding mode, and the presence of Y257. Steric interference due to different ICL loop structures may also interfere with G protein activation. No one thing has proven to confer ACKR3 with G protein activity including swapping all of the ICLs to those of canonical chemokine receptor, suggesting it is a combination of these different factors. The following has been added to the discussion on Pg. 13 to clearly note that any one feature is unlikely to drive the atypical behavior of ACKR3:

"The atypical activation of ACKR3 does not appear to be dependent on any singular receptor feature and is likely a combination of several factors."

> *(4) The extent to which conformational heterogeneity is a characteristic feature of ACKRs that contributes to their promiscuity and arrestin bias is unclear. The key residue the authors find promotes ACKR3 conformational heterogeneity is not conserved in most other ACKRs, but alternative mechanisms could generate similar heterogeneity.*

Despite the commonalities in the roles of the ACKRs, they all appear to have evolved independently. Thus, we do not believe that all features observed and described for one ACKR will explain the behavior of another. We have carefully avoided expanding our observations to other ACKRs to avoid suggesting common mechanisms.

> *(5) There are no data to confirm that the two receptors retain the same functional profiles observed in cell-based systems following in vitro manipulations (purification, labeling, nanodisc reconstitution).*

We agree this is an important point. All labeled receptors responded to agonist stimulation as expected. As only properly folded receptors are able to make the extensive interactions with ligands necessary for conformational changes (for instance, CXCL12 interacts with all TMs and ECLs), this suggests that the proteins are folded correctly and functional following all manipulations.

> ***Reviewer #3 (Public Review):***
>
> *Summary:*
>
> *This is a well-designed and rigorous comparative study of the conformational dynamics of two chemokine receptors, the canonical CXCR4 and the atypical ACKR3, using single-molecule fluorescence spectroscopy. These receptors play a role in cell migration and may be relevant for developing drugs targeting tumor growth in cancers. The authors use single-molecule FRET to obtain distributions of a specific intermolecular distance that*

> *changes upon activation of the receptor and track differences between the two receptors in the apo state, and in response to ligands and mutations. The picture emerging is that more dynamic conformations promote more basal activity and more promiscuous coupling of the receptor to effectors.*
>
> *Strengths:*
>
> *The study is well designed to test the main hypothesis, the sample preparation and the experiments conducted are sound and the data analysis is rigorous. The technique, smFRET, allows for the detection of several substates, even those that are rarely sampled, and it can provide a "connectivity map" by looking at the transition probabilities between states. The receptors are reconstituted in nanodiscs to create a native-like environment. The examples of raw donor/acceptor intensity traces and FRET traces look convincing and the data analysis is reliable to extract the sub-states of the ensemble. The role of specific residues in creating a more flat conformational landscape in ACKR3 (e.g., Y257 and the C34-C287 bridge) is well documented in the paper.*
>
> *Weaknesses:*
>
> *The kinetics side of the analysis is mentioned, but not described and discussed. I am not sure why since the data contains that information. For instance, it is not clear if greater conformational flexibility is accompanied by faster transitions between states or not.*

The reviewer is correct that kinetic information is available, in principle, from smFRET experiments. However, a detailed kinetic analysis will require a much larger data set than we currently possess, to adequately sample all possible transitions and the dwell times of each FRET state. We intend to perform such an analysis in the future as more data becomes available. The purpose of this initial study was to explore the conformational landscapes of CXCR4 and ACKR3 and to reveal differences between them. To this end, we have documented major differences in conformational preferences and response to ligands of the two receptors that are likely relevant to their different biological behavior. Future kinetic information will add further detail, but is not expected to alter the conclusions drawn here.

> *The method to choose the number of states seems reasonable, but the "similarity" of states argument (Figures S4 and S6) is not that clear.*

We thank the reviewer for noting a need for further clarification. We qualitatively compared the positions of the various FRET peaks across treatments to gain insight into the consistency of the conformations and avoid splitting real states by overfitting the data. For instance, fitting the ACKR3 treatments with three states leads to three distinct FRET populations for the R' intermediate. Adding a fourth state results in two intermediates that are fairly well overlapping. In contrast, the two-intermediate model for CXCR4 appears to split the R* state of the CXCL12 treated sample and causes a general shift in both intermediate states to lower FRET values when CXCL12 is present. As we assume that the conformations are consistent throughout the treatments, we conclude that this represents an overfitting artifact and not a novel CXCL12CXCR4 R*' state. Additional sentences have been added to the supplemental figure legend to better describe the comparative analysis.

"(Top) With the 3-state model, the R' states for apo-CXCR4 and for CXCL12- and IT1t-bound receptor overlapped well with similar apparent FRET values across all of the tested conditions. In the case of the four-state model, the R*' (Middle) and R' (Bottom) states were substantially different across the ligand treatments. In particular, the R*' state with CXCL12 treatment appears to arise from a splitting of the R* conformation, indicating that the model was overfitting the data."

> *Also, the "dynamics" explanation offered for ACKR3's failure to couple and activate G proteins is not very convincing. In other studies, it was shown that activation of GPCRs by agonists leads to an increase in local dynamics around the TM6 labelling site, but that did not prevent G protein coupling and activation.*

We agree with the reviewer that any single explanation for ACKR3 bias, including the dynamics argument presented here, is insufficient to fully characterize the ACKR3 responses. As noted by the reviewer, the TM6 movement and dynamics is generally correlated with G protein coupling, whereas other dynamics studies (Wingler et al. Cell 2019) have noted that arrestinbiased ligands do not lead to the same degree of TM6 movement. We have added the following statement to the discussion on Pg. 13:

"The atypical activation of ACKR3 does not appear to be dependent on any singular receptor feature and is likely a combination of several factors."

> ***Recommendations for the authors:***
>
> ***Reviewer #1 (Recommendations For The Authors):***
>
> *I would like to raise a technical point about the calculation and reporting of the FRET efficiency. The authors report the FRET efficiency as E=IA/(IA+ID). There is now a strong recommendation from the FRET community (https://doi.org/10.1038/s41592-018-0085-0) to use the term "FRET efficiency" only when a proper correction procedure of all correction factors has been applied, which is not the case here (gamma factor has not been calculated). The authors should therefore use the term "Apparent FRET Efficiency" and E_{app} in all the manuscripts.*
>
> *Also, it would be nice to indicate directly on the figures whether a ligand that is used is an agonist, antagonist, inverse agonist, etc...*

We thank the reviewer for suggesting this clarification in terminology. We now refer to apparent FRET efficiency (or E_{app}) throughout the manuscript and in the figures. In addition, we have added ligand descriptions to the relevant figures.

> ***Reviewer #2 (Recommendations For The Authors):***
>
> *(1) M159(4.40)C/Q245(6.28)C ACKR3 appears to have higher constitutive activity than ACKR3 Wt (Fig. S1). While the vehicle point itself is likely not significant due to the error in the Wt, the overall trend is clear and arguably even stronger than the effect of Y257(6.40)L (Fig. S9). While this is an inherent limitation of the method used, it should be clearly acknowledged; the comment in lines 162-164 seems to skirt the issue by only saying that arrestin recruitment is retained. It would be helpful and more rigorous to report the curve fit parameters (basal, E_{max}, EC50) for the arrestin recruitment experiments and the associated errors/significance (see https://www.graphpad.com/guides/prism/latest/statistics/stat_qa_multiple_comparisons_ after_.htm for a discussion).*

The Emin, E_{max}, and EC50 for M159$^{4.40}$C/Q245$^{6.28}$C ACKR3 were compared against the values for WT ACKR3 from Fig. S1 and only the E_{max} was determined to be significantly different by the extra sum of squares F test. A note has been added to the text to reflect these results on Pg. 5.

"Only the E_{max} for arrestin recruitment to CXCL12-stimulated ACKR3 was significantly altered by the mutations, while all other pharmacological parameters were the same as for WT receptors."

> *(2) The methods do not specify the reactive group of the dyes used for labeling (i.e., AlexaFluor 555-maleimide and Cy5-maleimide?).*

We regret the omission and have added the necessary details to the materials and methods.

> *(3) Were any of the native Cys residues removed from ACKR3 and CXCR4 in the constructs used for smFRET? ACKR3 appears to have two additional Cys residues in the N-terminus besides the one involved in the second disulfide bridge, and these would presumably be solvent-exposed. If so, please specify in the Methods and clarify whether the constructs tested in functional assays included these. (Also, please specify if the human receptors were used.)*

No additional cysteine residues were mutated in either receptor. All exposed cysteines are predicted to form disulfides. The residues in the N-terminus that the reviewer alludes to, C21 and C26, form a disulfide (Gustavsson et al. Nature Communications 2017) and are thus protected from our probes. Consistent with these expectations, neither WT CXCR4 nor ACKR3 exhibited significant fluorophore labeling (now mentioned in the text on Pg. 5). The species of origin has been added to the material and methods.

> *(4) There are a few instances where the data seem to slightly diverge from the proposed models that may be helpful to comment on explicitly in the text:*
>
> *- Figure 4E (ACKR3/CXCL12(P2G)): As noted in the legend, despite stabilizing R*/R*', CXCL12(P2G) reduces transitions between these states compared to Apo. This is more similar to the effects of VUF16840 (Figure 3D) than the other ACKR3 agonists. The authors note the difference between CXCL12(LHRQ) and CXCL12(P2G) (but not vs Apo) in this regard. There might be some other information here regarding the relative importance of the conformational equilibrium vs transition rates for receptor activity.*

Although the TDPs for CXCL12$_{P2G}$ and VUF16840 are similar, as noted by the reviewer, the overall FRET envelopes are drastically different.

The differences in transition probabilities for R ↔ R' and R*' « R* transitions observed in the presence of CXCL12$_{P2G}$ or CXCL12$_{LRHQ}$ relative to the apo receptor are now explicitly noted in the Results.

> *- The conformational distributions of ACKR3 apo and ACKR3 Y257L CXCL12 are very similar (Figure 5A,D). However, there is a substantial difference in the basal activity of WT vs CXCL12stimulated Y257L (Figure S9).*

The mutation Y257L appears to promote the highest and lowest FRET states at the expense of the intermediates. Although the distribution appears similar between Apo-WT and CXCL12Y257L, the depopulation of the R' state may lead to the observed activation in cells.

> *(5) There are inconsistent statements regarding the compatibility of G protein binding to the "active-like" ACKR3 conformation observed in the authors' previous structures (Yen et al, Sci Adv 2022). In the introduction, the authors seem to be making the case that steric clashes cannot account for its lack of coupling; in the discussion, they seem to consider it a possibility.*

The introduction to previous research on the molecular mechanisms governing the lack of ACKR3-G protein coupling was not intended to be all encompassing, but rather to highlight previous efforts to elucidate this process and justify our study of the role of dynamics. Due to the positions of the probes, we can only comment on the impact on TM6 movements and not other conformational changes. The steric clash reported in Yen et al. was in ICL2 and not directly tested here, so our observations do not preclude changes occurring in this region. We also do not claim that the active-like state resolved in our previous structures matches any specific state isolated here by smFRET.

> *(6) Line 83-85: "Having excluded other mechanisms we therefore surmised that the inability of ACKR3 to activate G proteins may be due to differences in receptor dynamics."*
>
> *Line 400-402: "It is possible that the active receptor conformation clashes sterically with the G protein as suggested by docking of G proteins to structures of ACKR3."*

As mentioned above, we suspect the mechanisms governing the inability of ACKR3 to couple to G proteins may be more complex than one particular feature but instead due to a combination of several factors. Accordingly, we have not completely eliminated a contribution of steric hindrance as we described in Yen et al. Sci Adv 2022 and instead include it as a possibility. Following the line highlighted here, we list several alternatives:

"Alternatively, the receptor dynamics and conformational transitions revealed here may prevent formation of productive contacts between ACKR3 and G protein that are required for coupling, even though G proteins appear to constitutively associate with the receptor."

And, at the end of the paragraph, we have added the following sentence:

"The atypical activation of ACKR3 does not appear to be dependent on any singular receptor feature and is likely a combination of several factors."

> *(7) If the authors believe that the various ligands/mutations are only altering the distribution/dynamics of the same 3/4 conformations of CXCR4/ACKR3, respectively, is there a reason each FRET efficiency histogram is fit independently instead of constraining the individual components to Gaussian components with the same centroids, and/or globally fitting all datasets for the same receptor?*

We performed global analysis across all data sets for each sample and condition. Since the peak positions of the various FRET states recovered in this way were consistent across treatments (Fig. S4,S6), we did not feel it was necessary to perform a further global analysis across all samples for a given receptor.

> ***Reviewer #3 (Recommendations For The Authors):***
>
> *The manuscript is well-written, the arguments are easy to follow and the figures are helpful and clear. Here are a few questions/suggestions that the authors might want to address before the paper will be published:*
>
> *(1) Include a table with kinetic rates between states in SI and have a brief discussion in the main text to support the trends observed in transition probabilities.*

As noted above, determining rate constants for each of the state-to-state transitions will require a much larger set of experimental smFRET data than is currently available and will be the subject of future studies.

> *(2) The argument of state similarity (Figure S4 and S6)... why are the profiles not Gaussian, like in the fits on Figures S3 and S5, repectively? I would also suggest that once the number of states is chosen to do a global fit, where the FRET values of a certain sub-state across different conditions for one receptor are shared.*

The state distributions presented in Figs. S4 and S6 (as well as throughout the rest of the paper) are derived from HMM fitting of the time traces themselves, and are not constrained to be Gaussian, whereas the GMM analysis in Figs. S3 and S5 are Gaussian fits to the final apparent FRET efficiency histograms.

Similar to our response to Review 2 above, due to the consistency of the fitted peak positions obtained across different conditions for a given sample, we did not feel that further global analysis was necessary.

> *(3) It is shown FRET changes from ~0.85 in the inactive (closed) state to ~0.25 in the active (open) state. How do these values match the expectations based on crystal structure and dye properties?*

As noted in our response to Reviewer 1, translating the apparent FRET values using the assumed Förster distances for A555/Cy5 (per FPbase) suggest a change in D-A distance of ~30 angstroms, whereas the expected change from structures is ~16 Å. We suspect this discrepancy is due to the lipids immediately adjacent to the fluorophores, which may lead to the probes being constrained in an extended position when TM6 moves outwards, thus also reporting the linker length in the distance change. Additionally, such interactions may constrain the donor and acceptor in unfavorable orientations for energy transfer, which would also reduce the FRET efficiency in the active state. Since the calculated D-A distance changes appear too large for GPCR activation, we have opted to not make any structural interpretations. Instead, all of our conclusions are based on resolving individual conformational states and quantifying their relative populations, which is based directly on the measured FRET efficiency distributions, not computed distances.

> *(4) The results on the effect of CXCL12-P2G on CXCR4 are confusing...despite being an antagonist, this ligand stabilizes the "active state"...I am not sure if the explanation offered is sufficient that the opening of the intracellular cleft is not sufficient to drive the G protein coupling/activation.*

We agree that the explanation related to the opening of the intracellular cleft being insufficient to drive G protein coupling/activation is speculative and we have removed that text. We now simply propose that the CXCL12 variants inhibit coupling of G proteins to CXCR4 or disrupt interactions necessary for signaling, as stated in the following text to the results on Pg. 8:

"Despite the ability of $CXCL12_{P2G}$ and $CXCL12_{LRHQ}$ to stabilize the active R* conformation of CXCR4, both variants are known to act as antagonists (20). This suggests that the CXCL12 mutants inhibit CXCR4 coupling to G proteins not by suppressing the active receptor population but rather by increasing the dynamics of the receptor state-to-state transitions. Our results suggest that the helical movements considered classic signatures of the active state may not be sufficient for CXCR4 to engage productively with G proteins."

https://doi.org/10.7554/eLife.100098.2.sa0